

Scope →

Single-Domain

Cross-Domain

Domain-Specific Model Optimization

FinSQL [1] — Finance Text-to-SQL

EHRSQL [2] — Healthcare Text-to-SQL

KorQuAD [3] — Korean MRC

Spider/BIRD [4,5] — General Text-to-SQL

Domain Pipeline Design (Single)

Self-RAG [12] — Retrieval architecture

RA-DIT [13] — Dual instruction tuning

(domain-specific pipelines, not generalized)

Cross-Domain Model Adaptation

Domain Adaptation Survey [29]

Fine-tuning / LoRA / Adapter methods

Multi-domain RAG [8,9]

(model weights, not architecture pattern)

*Model → Architecture
shift*

★ DKAP (This Work)

5-Layer Architecture Pattern

29 Systems · 16 Paradigms · 7 Industries
AI-Engine-Agnostic (LLM/RL/VLM/ML)

← **Research Gap Filled** →

Model-Centric

Architecture-Centric

Optimization Focus →